

MAZAR-I-SHARIF INTL, Afghanistan

WMO: 409110

Lat: 36.7055N

Lon: 67.2185E

Elev: 391

StdP: 96.71

Time Zone: 4.50 (AFG)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.0	-12.1	1.4	4.0	-9.9	1.7	10.1	11.9	8.4	10.8	7.9	2.6	80	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	41.8	18.7	40.8	18.4	39.2	17.9	19.5	38.0	19.0	36.8	18.6	36.0	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	11.5	20.6	14.2	10.6	20.5	13.2	9.9	20.7	57.0	36.9	55.3	36.6	53.9	36.1	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.2	9.8	8.7	DB	-8.1	43.9	5.5	0.6	-12.1	44.3	-15.3	44.6	-18.4	45.0	-22.4	45.4	
			WB	-8.5	20.7	5.1	0.6	-12.1	21.2	-15.1	21.6	-18.0	21.9	-21.7	22.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	5.9	6.8	13.5	19.4	26.7	31.7	34.3	32.1	27.0	19.2	11.0	6.4
	DBStd	10.96	3.61	5.47	4.61	4.39	3.06	2.66	2.25	2.73	3.11	4.41	4.83	4.23
	HDD10.0	427	132	109	21	1	0	0	0	0	0	1	40	122
	HDD18.3	1543	384	323	158	41	0	0	0	0	1	44	224	368
	CDD10.0	3929	6	20	130	283	518	652	754	686	511	287	71	12
	CDD18.3	2001	0	0	9	72	259	402	496	427	261	71	4	0
	CDH23.3	30191	0	2	102	670	3412	6535	8569	6898	3337	624	40	0
	CDH26.7	18826	0	0	17	222	1781	4289	6104	4521	1687	197	9	0
Wind	WSAvg	3.6	3.0	3.5	3.8	3.6	3.6	4.1	4.6	4.2	3.4	3.1	2.8	2.9
Precipitation	PrecAvg	179	34	35	47	24	11	0	0	0	0	5	11	22
	PrecMax	241	67	81	143	55	41	3	0	1	8	25	55	107
	PrecMin	133	1	6	6	0	0	0	0	0	0	0	0	2
	PrecStd	28	19	18	29	15	11	1	0	0	2	7	12	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	22.8	29.0	34.2	39.1	42.1	43.2	42.0	38.8	33.9	28.1	19.2
		MCWB	9.1	12.7	15.1	16.0	17.4	18.1	19.2	19.3	17.4	14.5	13.2	10.1
	2%	DB	15.2	18.2	26.1	31.2	37.1	40.9	42.2	40.8	37.0	31.0	23.2	16.9
		MCWB	8.8	11.2	14.4	16.3	17.3	18.1	18.8	18.5	16.7	14.7	11.6	8.7
	5%	DB	13.8	16.1	23.9	29.1	35.8	39.8	41.1	39.2	35.2	28.9	20.8	14.9
		MCWB	8.1	10.4	13.9	15.6	16.9	17.8	18.6	17.9	16.2	14.0	11.2	8.7
	10%	DB	12.0	14.1	21.2	27.1	34.0	38.1	40.0	38.1	33.9	26.9	18.2	12.9
		MCWB	7.6	9.6	13.3	15.6	16.7	17.3	18.3	17.5	15.8	13.4	10.3	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	13.9	16.8	19.3	19.5	19.4	20.5	20.0	18.4	16.6	14.9	12.5
		MCDB	15.0	18.2	24.7	26.7	29.5	38.3	39.7	40.0	35.6	28.8	22.7	16.1
	2%	WB	10.0	12.4	15.7	18.3	18.7	18.9	19.7	19.1	17.5	15.6	13.7	10.4
		MCDB	13.1	16.8	22.7	25.8	30.4	37.5	40.2	38.4	34.5	27.3	19.9	14.7
	5%	WB	9.2	11.1	14.7	17.5	18.3	18.5	19.1	18.5	16.8	14.8	12.3	9.4
		MCDB	12.6	15.3	21.7	24.9	29.9	36.7	39.4	37.8	33.8	26.2	18.4	12.8
	10%	WB	8.0	10.0	13.9	16.7	17.7	18.0	18.6	17.9	16.1	14.0	11.0	8.3
		MCDB	11.4	13.7	20.4	24.2	29.5	35.6	38.7	37.1	32.7	25.4	17.3	11.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.4	9.9	10.3	11.9	10.9	9.9	9.9	11.1	10.7	9.3	8.4
		MCDBR	10.7	11.7	13.9	13.5	14.1	12.7	11.4	11.2	13.3	13.3	13.4	11.9
		MCWBR	6.1	6.0	5.4	4.7	4.8	4.5	4.4	4.4	5.6	5.8	6.4	6.6
	5% WB	MCDBR	9.1	9.7	12.3	11.1	11.9	11.1	10.0	10.0	12.0	11.3	11.1	9.7
	MCWBR	5.7	5.5	5.5	4.3	3.8	4.1	4.2	4.5	5.4	5.2	5.9	6.1	
Clear-Sky Solar Irradiance	taub	0.379	0.434	0.490	0.507	0.506	0.509	0.494	0.469	0.440	0.465	0.439	0.386	
	taud	2.112	1.968	1.858	1.861	1.882	1.864	1.896	1.958	1.993	1.899	1.954	2.086	
	Ebn at Noon	778	775	768	784	791	787	795	807	806	729	695	735	
	Edh at Noon	119	155	190	200	199	202	194	179	163	164	138	115	
All-Sky Solar Radiation	RadAvg	2.27	3.04	4.40	5.69	7.11	8.00	7.85	7.17	6.09	4.39	2.81	2.11	
	RadStd	0.21	0.35	0.34	0.51	0.37	0.21	0.17	0.21	0.12	0.26	0.27	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.54	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page